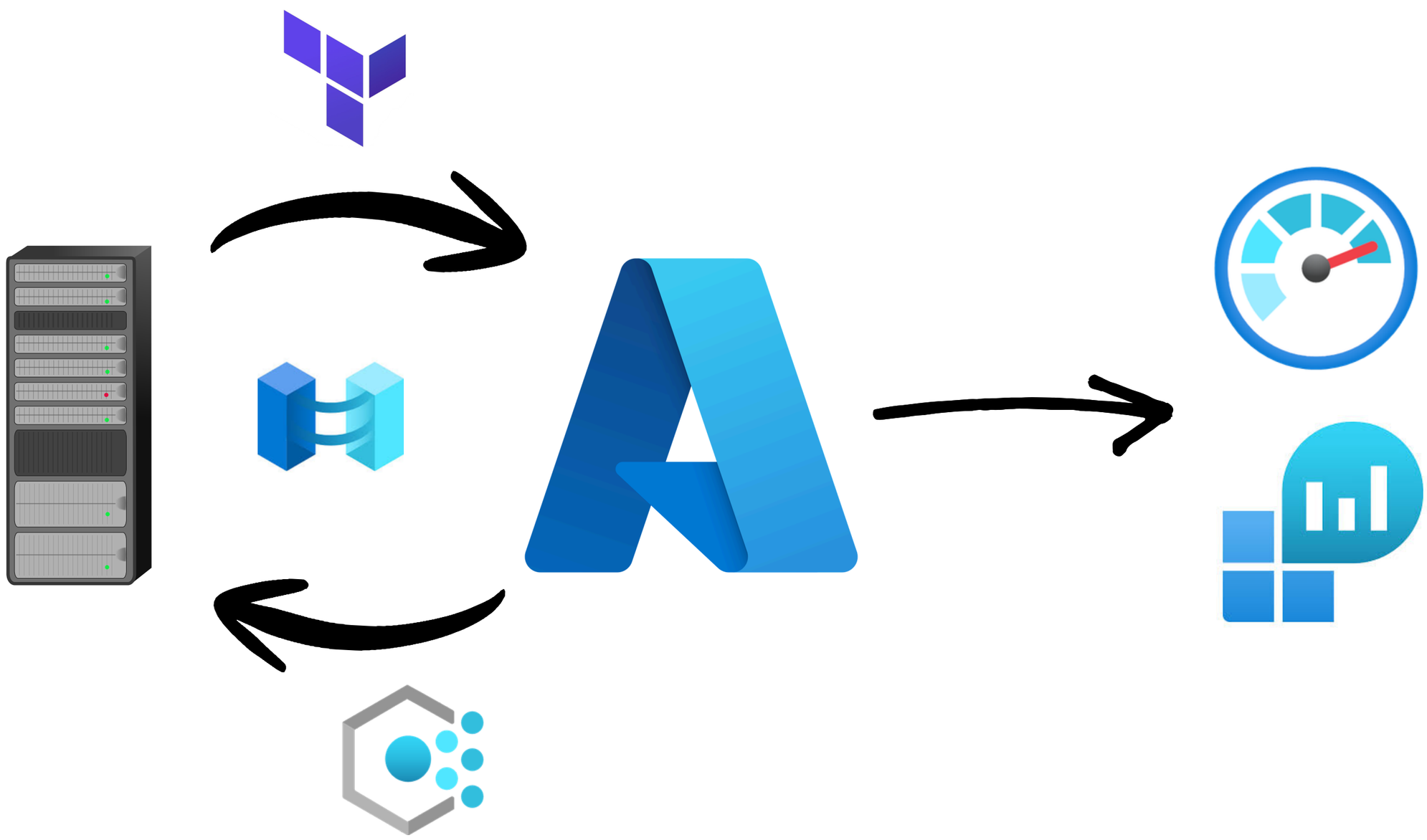
PROJECT DOCUMENTATION

Hybrid Cloud Governance & IaC Automation via Azure Arc



Project made by Ayoub Anlouf

Table of contents

I - Creating the local “On-Prem” Environment

1 - Creating the VMware Workstation Virtual Machine

2 - Installing the Operating System on the VM (Ubuntu)

II - Installing the prerequesites

1 - Installing Azure CLI & Terraform

III - Creating the Cloud environment with Terraform (IaC)

1 - Provider definition

2 - Defining the variables

3 - Creating the Resource Group, Application and Service Principal

4 - Defining the Service Principal credentials outputs

5 - Applying the Infrastructure configuration

IV - Connecting the “On-Prem” server to the Cloud

1 - Registering the required Azure Resource Providers

2 - Installing AZCMAgent on the Virtual Machine

3 - Retrieving the Service Principal Credentials

4 - Connecting the Virtual Machine to Azure Arc

V - Post-Deployment Governance

1 - Policy Assignment via Terraform

2 - Enabling Azure Extensions for Monitoring

VI - Repository Publishing

1 - Creating a .gitignore file

2 - Pushing the files to the GitHub repository

Links

VMware Workstation - https://www.vmware.com/products/desktop...

Ubuntu - https://ubuntu.com/download/desktop

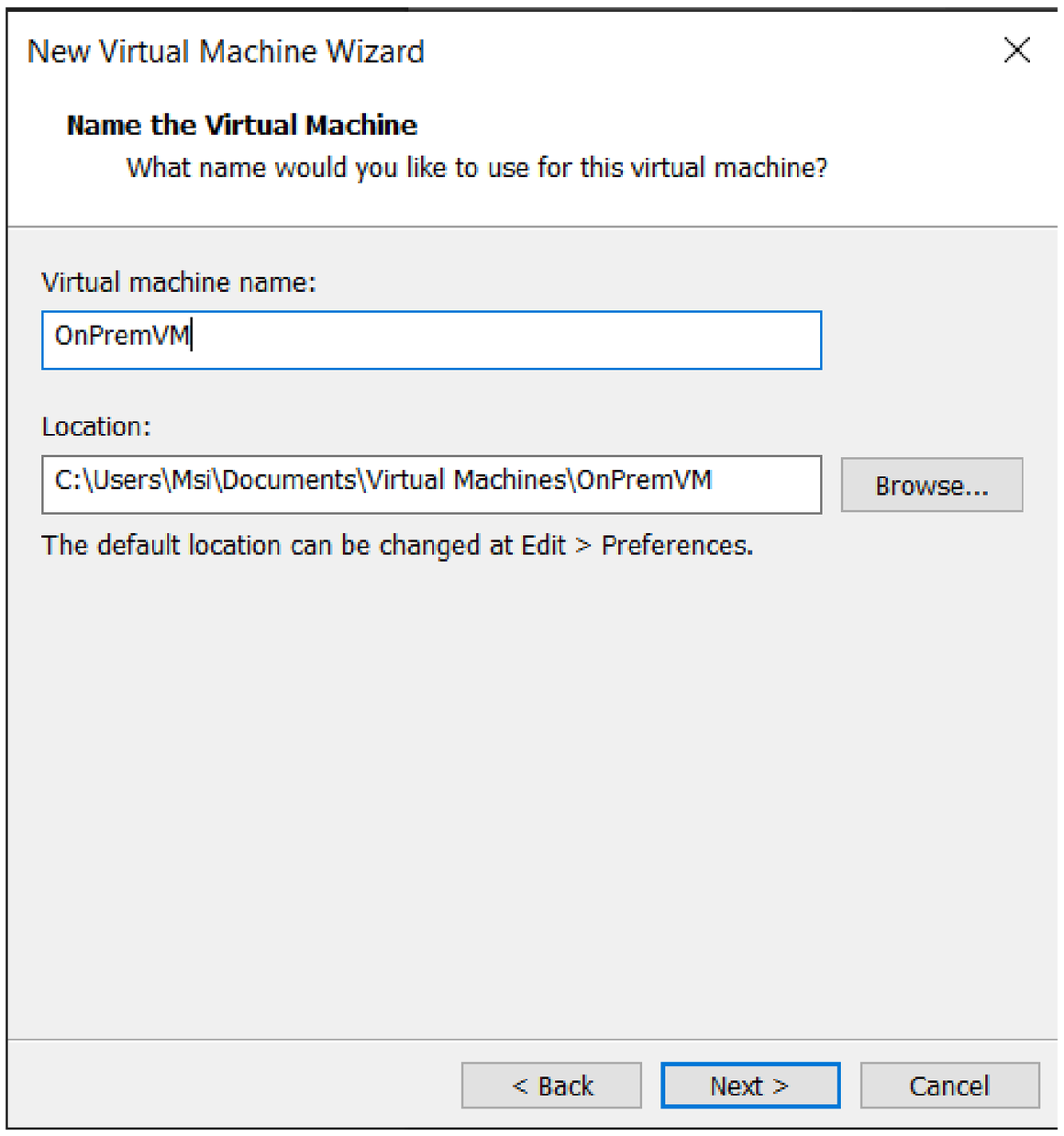
Azure CLI - https://learn.microsoft.com/en-us/cli/azure/install-azur...

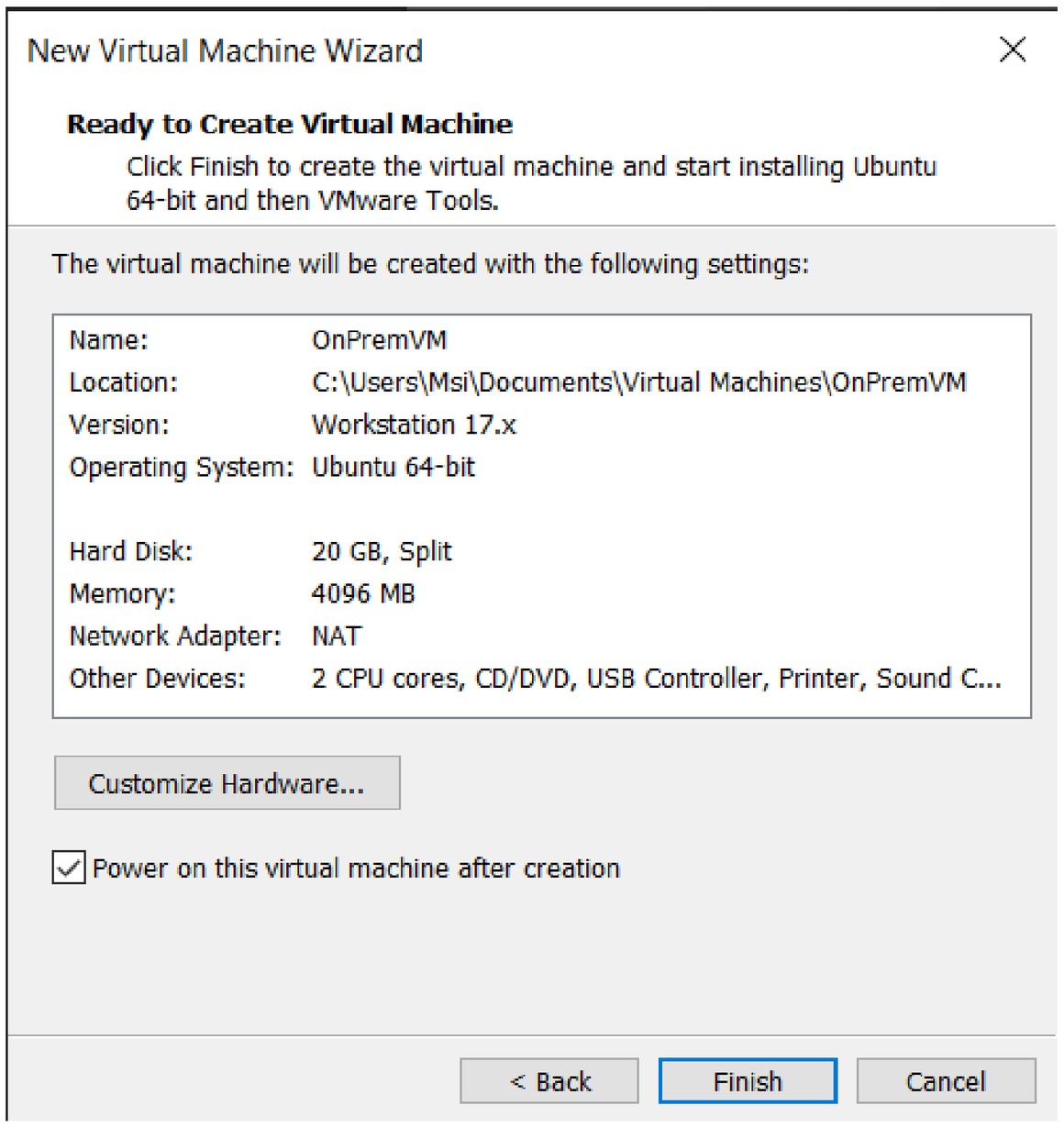
Terraform - https://developer.hashicorp.com/terraform/install

Git - https://git-scm.com/install/

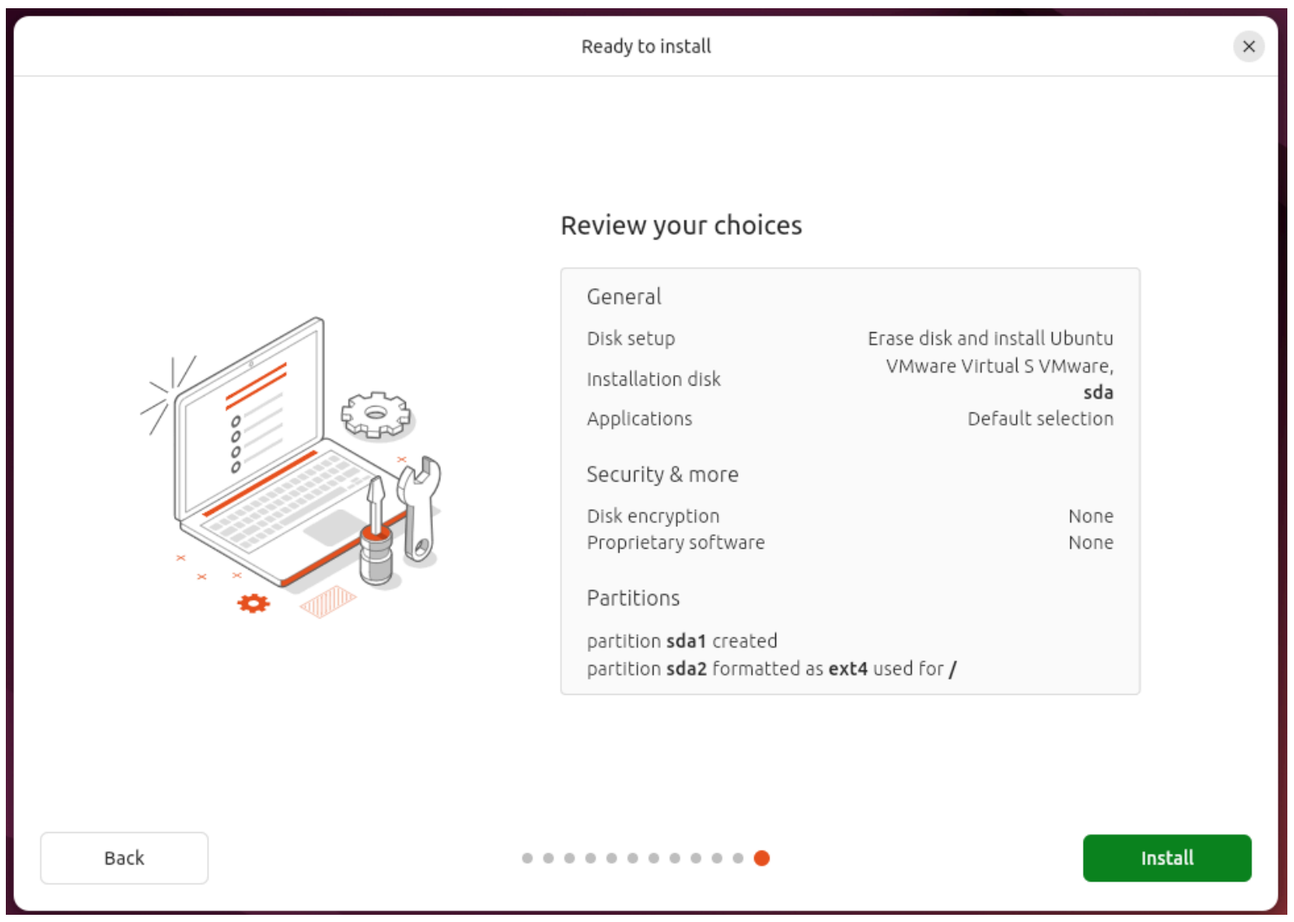
I - Creating the local “On-Prem” Environment

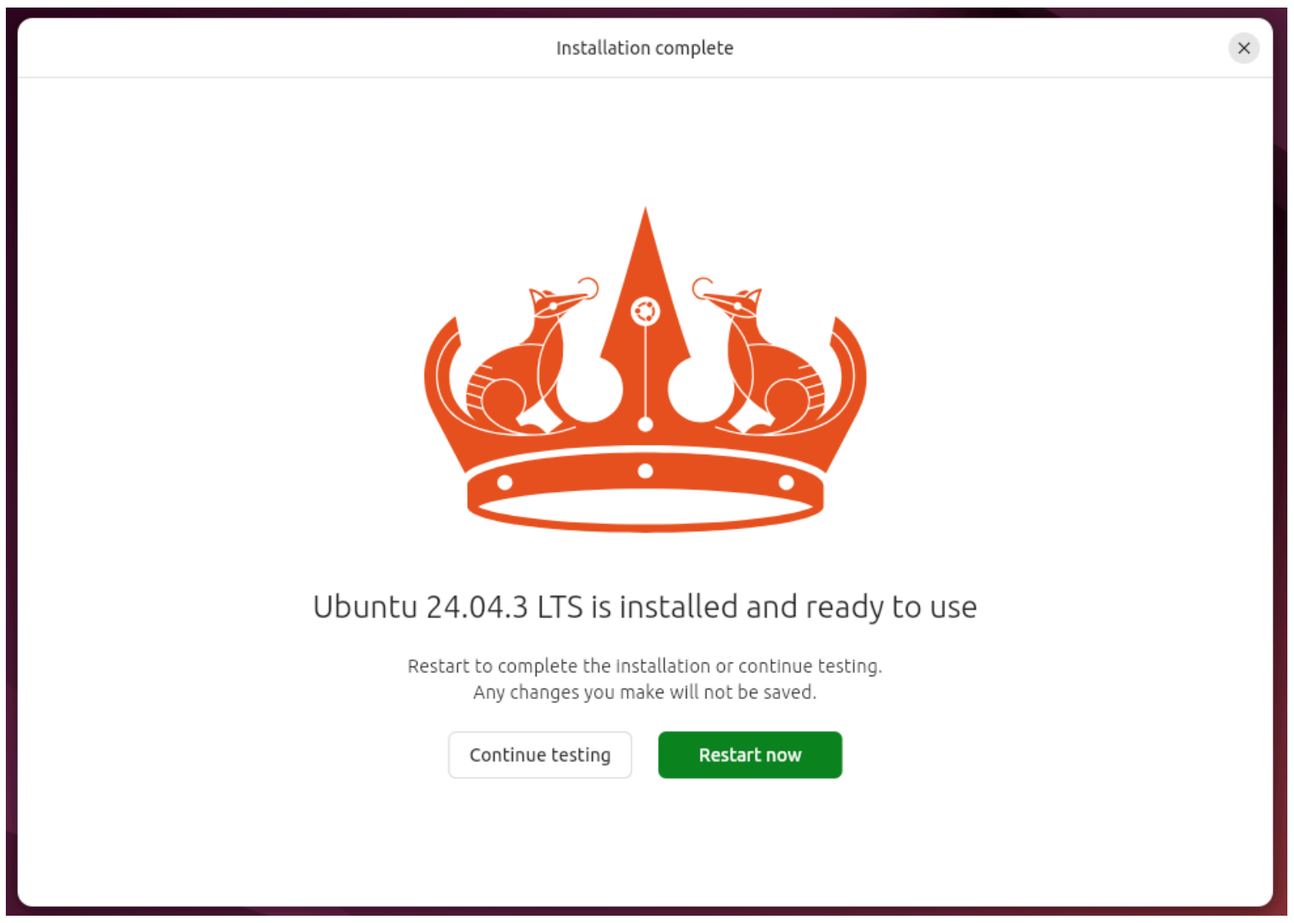
1 - Creating the VMware Workstation Virtual Machine





2 - Installing the Operating System on the VM (Ubuntu)

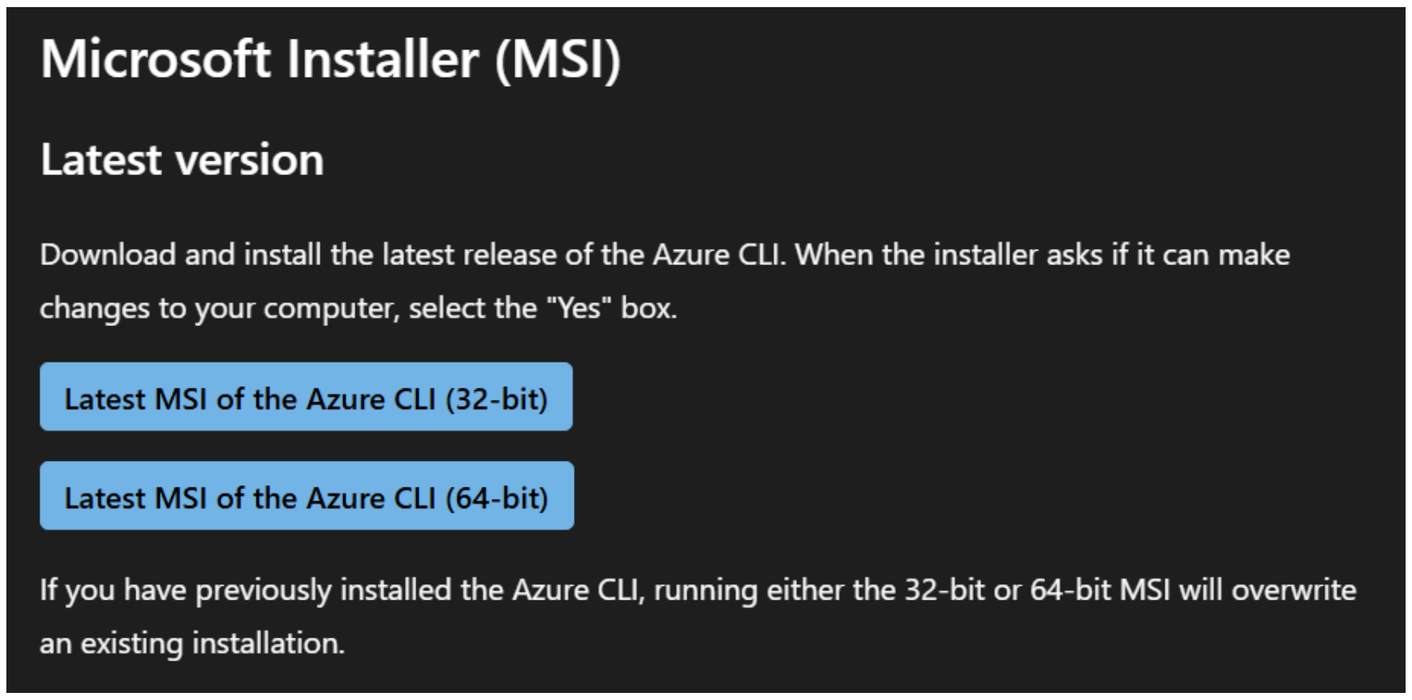


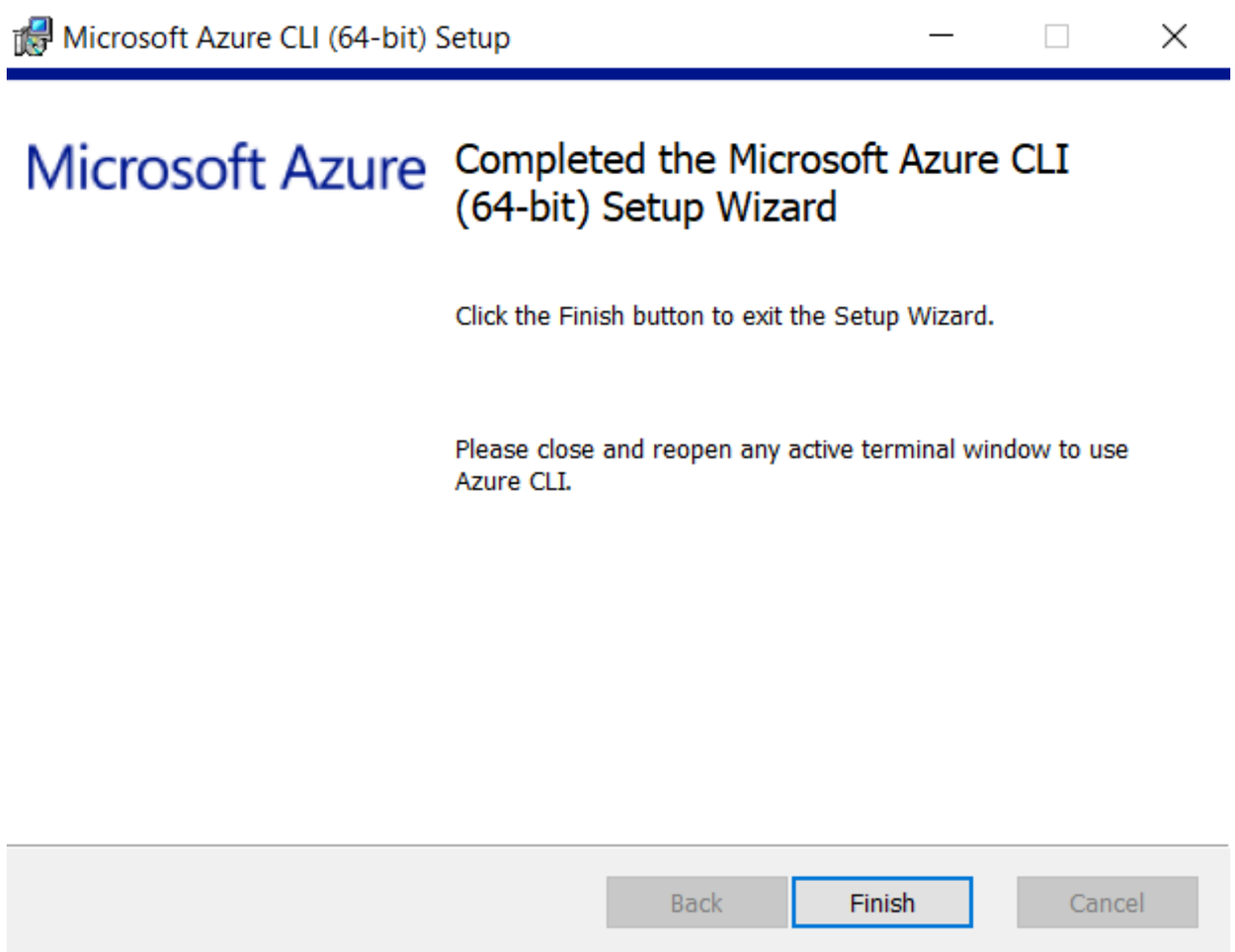


II - Installing the prerequesites

1 - Installing Azure CLI & Terraform

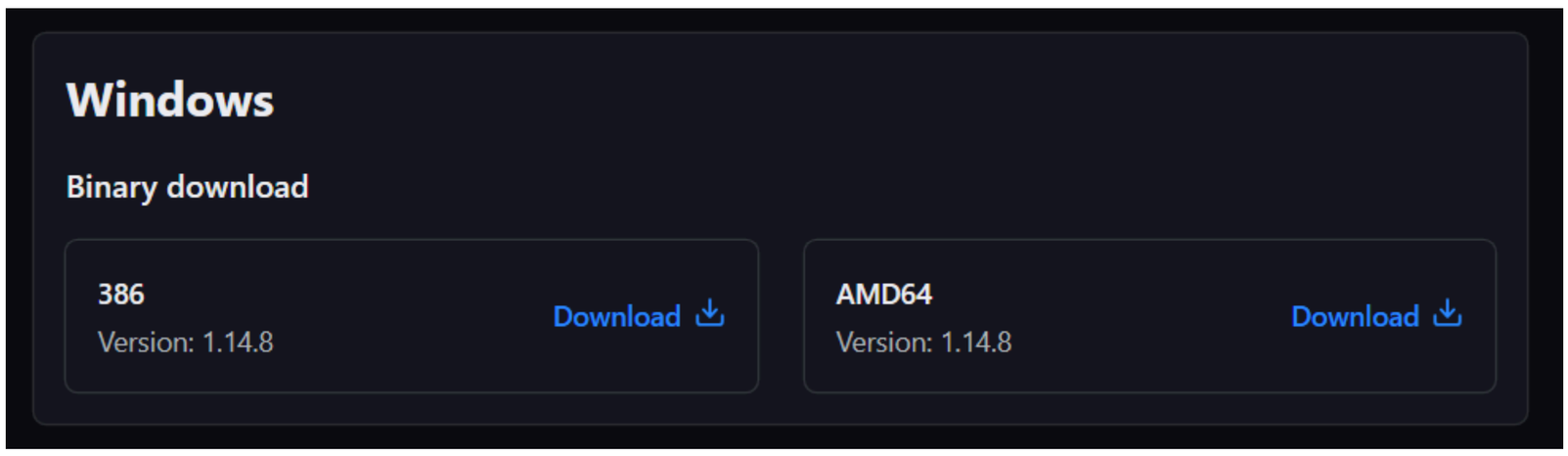
Installing Azure CLI

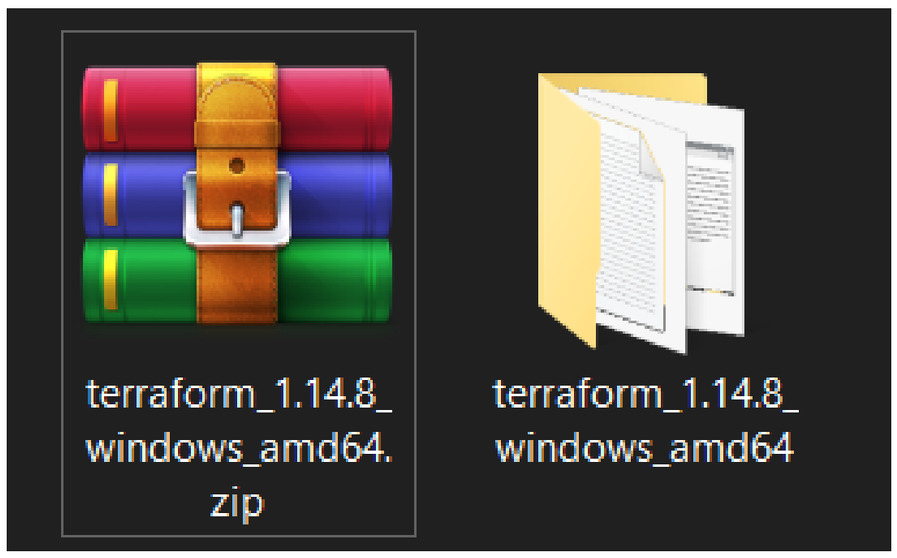


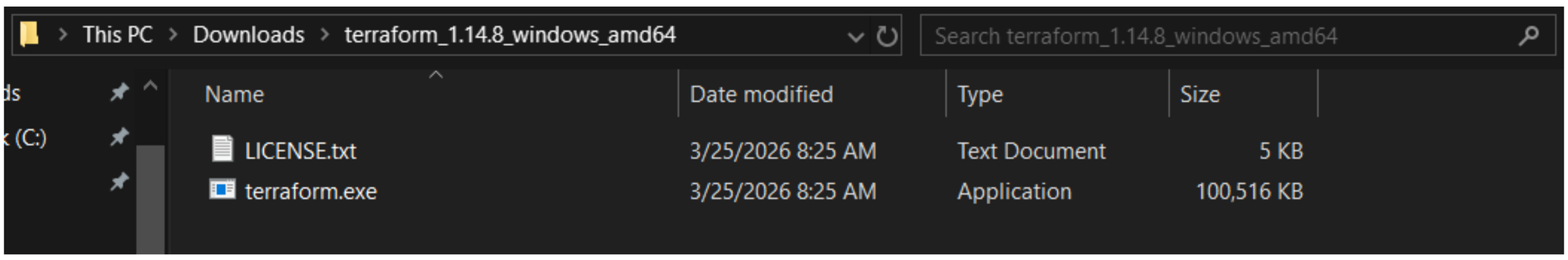


```
PS C:\Users\Msi\Desktop\AzureArc> az --version
azure-cli                         2.85.0

core                              2.85.0
telemetry                          1.1.0

Dependencies:
msal                              1.35.1
azure-mgmt-resource               24.0.0

Python location 'C:\Program Files\Microsoft SDKs\Azure\CLI2\python.exe'
Config directory 'C:\Users\Msi\.azure'
Extensions directory 'C:\Users\Msi\.azure\cliextensions'

Python (Windows) 3.13.11 (tags/v3.13.11:6278944, Dec  5 2025, 16:26:58) [MSC v.1944 64 bit (AMD64)]

Legal docs and information: aka.ms/AzureCliLegal


Your CLI is up-to-date.
```

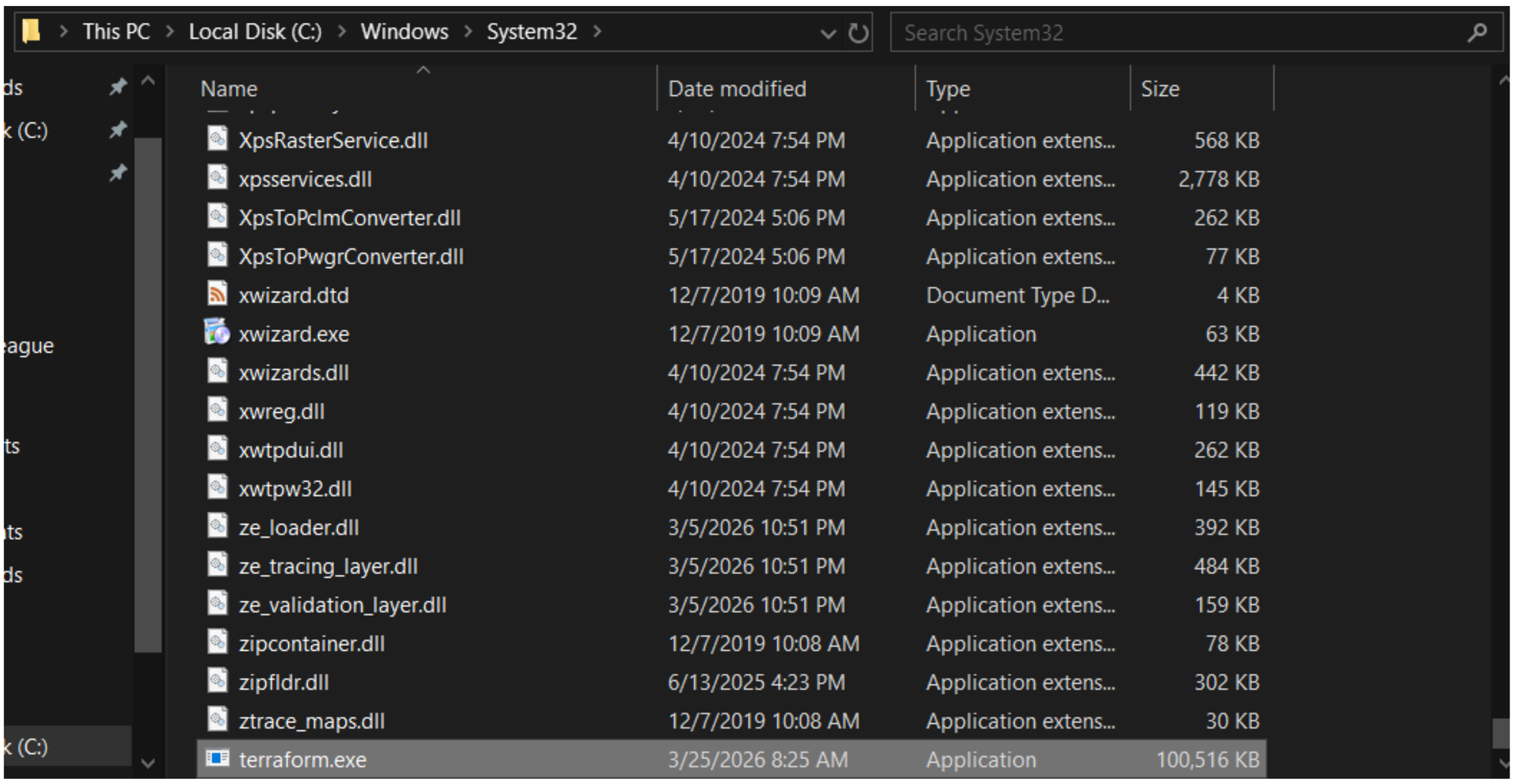
Installing Terraform







Moving the file terraform.exe to Windows/System32




```
PS C:\Users\Msi\Desktop\AzureArc> terraform --version
Terraform v1.14.8
on windows_amd64
PS C:\Users\Msi\Desktop\AzureArc>
```

III - Creating the Cloud environment with Terraform (IaC)

1 - Provider definition

provider.tf

```
provider "azurerm" {
  features {}
  subscription_id = var.subscription_id
  client_id       = var.client_id
  client_secret   = var.client_secret
  tenant_id       = var.tenant_id
}
```

2 - Defining the variables

variables.tf

```
variable "subscription_id" {}
variable "client_id" {}
variable "client_secret" {}
variable "tenant_id" {}
```

3 - Creating the Resource Group, Application and Service Principal

main.tf

```
resource "azurerm_resource_group" "rg" {
  name     = "ArcProjectRG"
  location = "westeurope"
}

resource "azuread_application" "arc_app" {
  display_name = "ArcProjectApp"
}

resource "azuread_service_principal" "arc_sp" {
  client_id = azuread_application.arc_app.client_id
}

resource "azuread_service_principal_password" "arc_sp_pwd" {
  service_principal_id = azuread_service_principal.arc_sp.id
  end_date             = timeadd(timestamp(), "8760h")
}

resource "azurerm_role_assignment" "arc_role" {
  scope                = azurerm_resource_group.rg.id
  role_definition_name = "Azure Connected Machine Onboarding"
  principal_id         = azuread_service_principal.arc_sp.object_id
}

```

4 - Defining the Service Principal credentials outputs

outputs.tf

```
output "sp_client_id" {
  value = azuread_service_principal.arc_sp.client_id
}

output "sp_client_secret" {
  value = azuread_service_principal_password.arc_sp_pwd.value
  sensitive = true
}

output "tenant_id" {
  value = var.tenant_id
}

```

5 - Applying the Infrastructure configuration

```
PS C:\Users\Msi\Desktop\AzureArc> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/azuread...
- Finding latest version of hashicorp/azurerm...
- Installing hashicorp/azuread v3.8.0...
- Installed hashicorp/azuread v3.8.0 (signed by HashiCorp)
- Installing hashicorp/azurerm v4.68.0...
- Installed hashicorp/azurerm v4.68.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\Msi\Desktop\AzureArc>
```

```
PS C:\Users\Msi\Desktop\AzureArc> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # azuread_application.arc_app will be created
  + resource "azuread_application" "arc_app" {
      + app_role_ids                = (known after apply)
      + client_id                   = (known after apply)
      + disabled_by_microsoft       = (known after apply)
      + display_name                = "ArcProjectApp"
      + id                          = (known after apply)
      + logo_url                    = (known after apply)
      + oauth2_permission_scope_ids = (known after apply)
      + object_id                   = (known after apply)
      + prevent_duplicate_names     = false
      + publisher_domain            = (known after apply)
      + sign_in_audience            = "AzureADMyOrg"
      + tags                        = (known after apply)
      + template_id                 = (known after apply)

      + feature_tags (known after apply)
    }

  # azuread_service_principal.arc_sp will be created
  + resource "azuread_service_principal" "arc_sp" {
      + account_enabled             = true
      + app_role_ids                = (known after apply)
      + app_roles                   = (known after apply)
      + application_tenant_id       = (known after apply)
      + client_id                   = (known after apply)
      + display_name                = (known after apply)
      + homepage_url                = (known after apply)
      + id                          = (known after apply)
      + logout_url                  = (known after apply)
      + oauth2_permission_scope_ids = (known after apply)
      + oauth2_permission_scopes    = (known after apply)
      + object_id                   = (known after apply)
      + redirect_uris               = (known after apply)
```

```
PS C:\Users\Msi\Desktop\AzureArc> terraform apply
azuread_application.arc_app: Refreshing state... [id=/applications/727c1f5e-1442-489c-a6f3-c8afbef85e4e]
azuread_service_principal.arc_sp: Refreshing state... [id=/servicePrincipals/79e9d074-4bab-465d-ab98-b329376e4eb1]
azuread_service_principal_password.arc_sp_pwd: Refreshing state... [id=79e9d074-4bab-465d-ab98-b329376e4eb1/password/d14c33ef-3f17-4c5f-b20c-33d53c4a68ce]
azurerm_resource_group.rg: Refreshing state... [id=/subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG]
```

```
Apply complete! Resources: 2 added, 0 changed, 1 destroyed.

Outputs:

sp_client_id = "62ea6520-b116-4330-9deb-c63a51e6029d"
sp_client_secret = <sensitive>
tenant_id = "631775c0-186e-4483-a8c5-f10afea80f5f"
PS C:\Users\Msi\Desktop\AzureArc>
```

IV - Connecting the “On-Prem” server to the Cloud

1 - Registering the required Azure Resource Providers

```
PS C:\Users\Msi\Desktop\AzureArc> az provider register --namespace Microsoft.HybridCompute
>> az provider register --namespace Microsoft.GuestConfiguration
>> az provider register --namespace Microsoft.HybridConnectivity
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.HybridCompute'
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.HybridConnectivity'
```

```
PS C:\Users\Msi\Desktop\AzureArc> az provider show --namespace Microsoft.HybridCompute --query "registrationState"
>> az provider show --namespace Microsoft.GuestConfiguration --query "registrationState"
>> az provider show --namespace Microsoft.HybridConnectivity --query "registrationState"
>>
"Registered"
"Registered"
"Registered"
PS C:\Users\Msi\Desktop\AzureArc>
```

2 - Installing AZCMAgent on the Virtual Machine

```
vif@vif:~$ wget https://aka.ms/azcmagent -O install_linux_azmagent.sh
--2026-04-15 17:05:45--  https://aka.ms/azcmagent
Resolving aka.ms (aka.ms)... 2.22.85.152, 64:ff9b::216:5598
Connecting to aka.ms (aka.ms)|2.22.85.152|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://gbl.his.arc.azure.com/azcmagent-linux [following]
--2026-04-15 17:05:46--  https://gbl.his.arc.azure.com/azcmagent-linux
Resolving gbl.his.arc.azure.com (gbl.his.arc.azure.com)... 172.202.65.10, 172.20
2.64.10, 2603:1030:13:201::10, ...
Connecting to gbl.his.arc.azure.com (gbl.his.arc.azure.com)|172.202.65.10|:443..
. connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/plain]
Saving to: ‘install_linux_azmagent.sh’

install_linux_azmag     [    <=>                 ]  40.99K  79.3KB/s    in 0.5s

2026-04-15 17:05:47 (79.3 KB/s) - ‘install_linux_azmagent.sh’ saved [41971]

vif@vif:~$ chmod +x install_linux_azmagent.sh
vif@vif:~$ sudo ./install_linux_azmagent.sh
```

```
Processing triggers for man-db (2.12.0-4build2) ...
Latest version of azcmagent is installed.
vif@vif:~$ azcmagent version
azcmagent version  1.63.03384.1014
vif@vif:~$
```

3 - Retrieving the Service Principal Credentials

```
PS C:\Users\Msi\Desktop\AzureArc> terraform output -json
{
  "sp_client_id": {
    "sensitive": false,
    "type": "string",
    "value": "62ea6520-b116-4330-9deb-c63a51e6029d"
  },
  "sp_client_secret": {
    "sensitive": true,
    "type": "string",
    "value": [redacted]
  },
  "tenant_id": {
    "sensitive": false,
    "type": "string",
    "value": "631775c0-186e-4483-a8c5-f10afea80f5f"
  }
}
```

4 - Connecting the Virtual Machine to Azure Arc

```
vif@vif:~$ sudo azcmagent connect   --service-principal-id 62ea6520-b116-4330-9deb-c63a51e6029d   --service-principal-se
cret ZAz8Q~yx~iH8LSjU.2f0Pqru5KvP-Wsc9RXTScUW   --resource-group ArcProjectRG   --tenant-id 631775c0-186e-4483-a8c5-f10a
fea80f5f   --subscription-id afd82d75-3dfb-4266-badc-e0c5afd8b8d1   --location westeurope
[sudo] password for vif:
INFO    Connecting machine to Azure... This might take a few minutes.
INFO    Cloud: AzureCloud
INFO    Testing connectivity to endpoints that are needed to connect to Azure... This might take a few minutes.
  20% [==>            ]
  30% [===>           ]
  INFO    Creating resource in Azure...                 Correlation ID=8238513a-1452-4c92-be43-2bf72a07065b Resource ID=
/subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG/providers/Microsoft.HybridCompute/machin
es/vif
INFO    Creating resource in Azure...                 Correlation ID=8238513a-1452-4c92-be43-2bf72a07065b Resource ID=/s
ubscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG/providers/Microsoft.HybridCompute/machines
/vif
  60% [========>      ]
  INFO    Retrieving MSI certificate... This might take a few minutes.
  80% [===========>   ]
 100% [===============]
  INFO    Connected machine to Azure
INFO    Machine overview page: https://portal.azure.com/#@631775c0-186e-4483-a8c5-f10afea80f5f/resource/subscriptions/af
d82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG/providers/Microsoft.HybridCompute/machines/vif/overview
vif@vif:~$
```

```
vif@vif:~$ azcmagent show
Resource Name                         : vif
Resource Group Name                   : ArcProjectRG
Resource Namespace                    : Microsoft.HybridCompute
Resource Id                           : /subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG/providers/Microsoft.HybridCompute/machines/vif
Subscription ID                       : afd82d75-3dfb-4266-badc-e0c5afd8b8d1
Tenant ID                             : 631775c0-186e-4483-a8c5-f10afea80f5f
VM ID                                 : 5bf9d25b-1908-47f6-917b-92c6d8231946
Correlation ID                        : 8238513a-1452-4c92-be43-2bf72a07065b
VM UUID                               : ac7c4d56-a605-4407-8831-54602b26668b
Location                              : westeurope
Cloud                                 : AzureCloud
Identity Key Store                    : Default
Agent Version                         : 1.63.03384.1014
Agent Logfile                         : /var/opt/azcmagent/log/himds.log
Agent Status                          : Connected
Agent Last Heartbeat                  : 2026-04-15T20:02:49+01:00
Agent Error Code                      :
Agent Error Details                   :
Agent Error Timestamp                 :
Using HTTPS Proxy                     :
Proxy Bypass List                     :
Upstream Proxy                        :
Gateway URL                           :
Cloud Provider                        : N/A
Cloud Metadata
Manufacturer                          : VMware, Inc.
Model                                 : VMware Virtual Platform
MSSQL Server Detected                 : false
MySQL Server Detected                 : false
PGSQL Server Detected                 : false
Dependent Service Status
  Agent Service (himdsd)              : active
  Azure Arc Proxy (arcproxyd)         : active
  Extension Service (extd)            : active
  GC Service (gcad)                   : active
Portal Page                           : https://portal.azure.com/#@631775c0-186e-4483-a8c5-f10afea80f5f/resource/subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resou
rceGroups/ArcProjectRG/providers/Microsoft.HybridCompute/machines/vif/overview
Disabled Features                     :
Agent Auto Upgrade Task Status        : enabled, id: /etc/cron.d/azcmagent_autoupgrade

vif@vif:~$
```

V - Post-Deployment Governance

1 - Policy Assignment via Terraform

policy.tf × terraform.tfvars

policy.tf

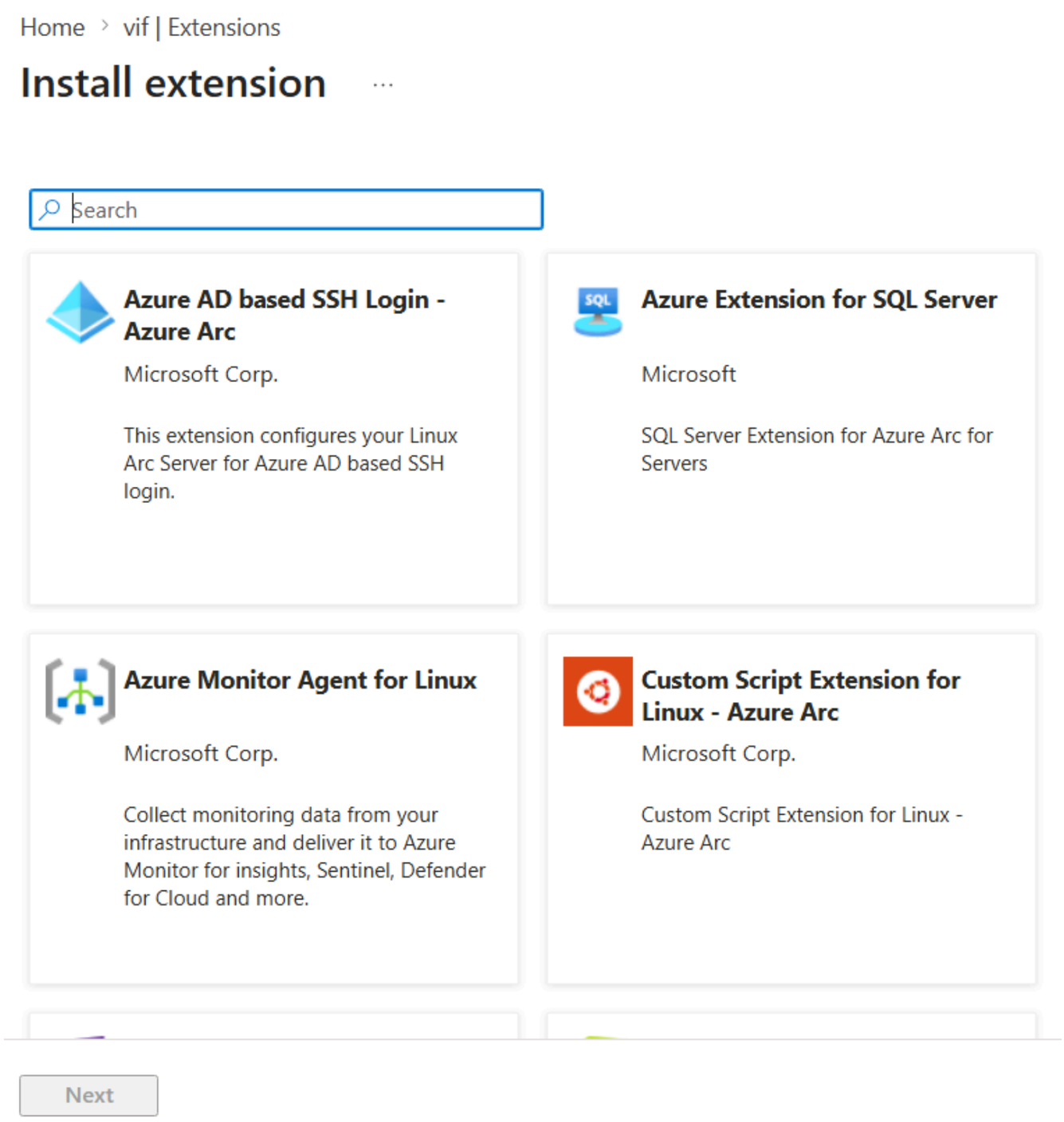
```
resource "azurerm_policy_definition" "arc_linux_ama" {
  name         = "ArcLinuxAMA"
  policy_type  = "Custom"       # BuiltIn policies cannot be recreated; use Custom here
  mode         = "Indexed"
  display_name = "Linux Arc-enabled machines should have Azure Monitor Agent installed"

  policy_rule = <<POLICY_RULE
{
  "if": {
    "allOf": [
      {
        "equals": "Microsoft.HybridCompute/machines",
        "field": "type"
      },
      {
        "equals": "linux",
        "field": "Microsoft.HybridCompute/machines/osName"
      }
    ]
  },
  "then": {
    "effect": "AuditIfNotExists",
    "details": {
      "type": "Microsoft.HybridCompute/machines/extensions",
      "existenceCondition": {
        "allOf": [
          {
            "equals": "AzureMonitorLinuxAgent",
            "field": "Microsoft.HybridCompute/machines/extensions/type"
          },
          {
            "equals": "Microsoft.Azure.Monitor",
            "field": "Microsoft.HybridCompute/machines/extensions/publisher"
          },
          {
            "equals": "Succeeded",
            "field": "Microsoft.HybridCompute/machines/extensions/provisioningState"
          }
        ]
      }
    }
  }
}
POLICY_RULE

}

resource "azurerm_resource_group_policy_assignment" "arc_linux_ama_assignment" {
  name                 = "ArcLinuxAMA"
  resource_group_id    = azurerm_resource_group.rg.id
  policy_definition_id = azurerm_policy_definition.arc_linux_ama.id
}

```

Reapplying the Terraform config

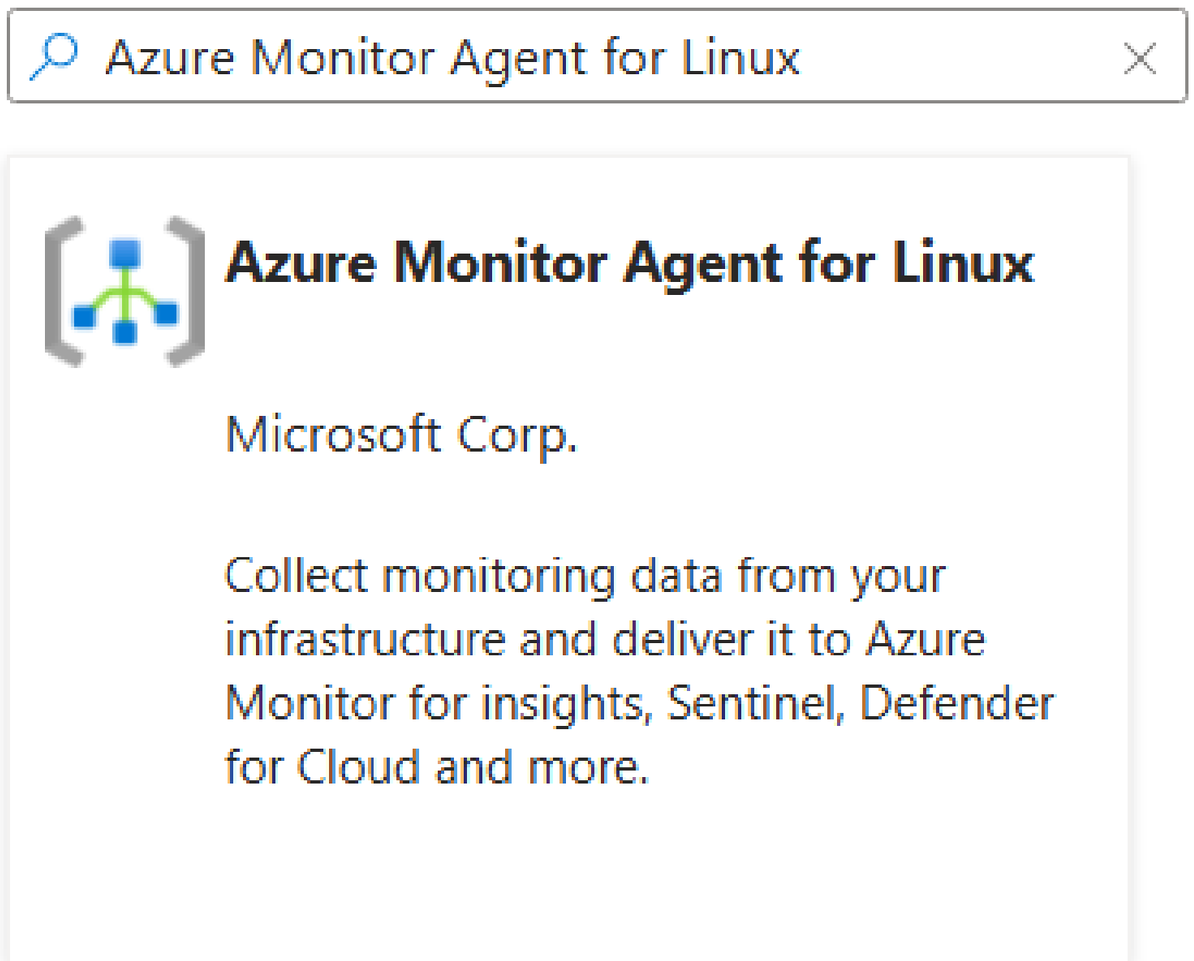
```
Apply complete! Resources: 3 added, 0 changed, 1 destroyed.

Outputs:

sp_client_id = "62ea6520-b116-4330-9deb-c63a51e6029d"
sp_client_secret = <sensitive>
tenant_id = "631775c0-186e-4483-a8c5-f10afea80f5f"
PS C:\Users\Msi\Desktop\AzureArc>
```

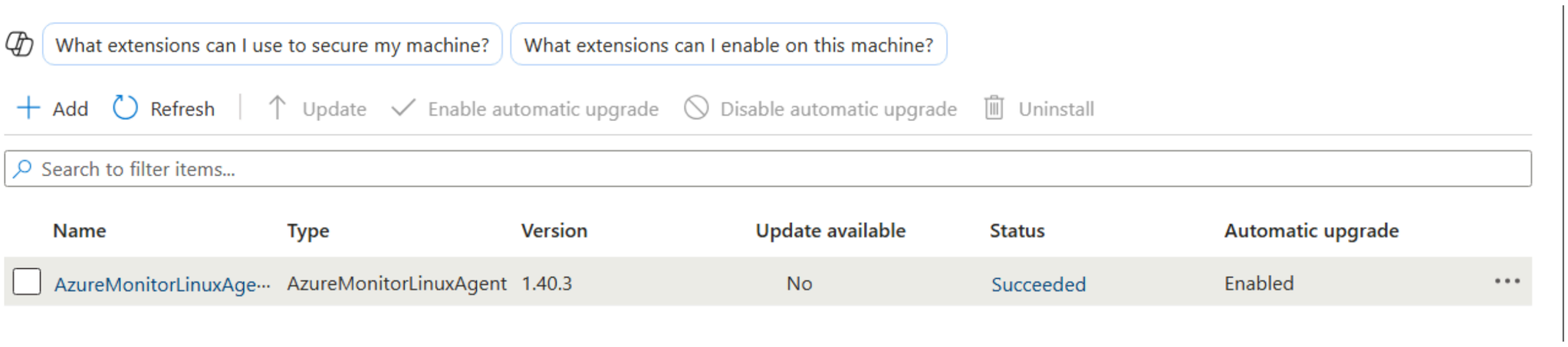
```
PS C:\Users\Msi\Desktop\AzureArc> az policy assignment list --scope "/subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG" --output table
DefinitionVersion    DisplayName    EnforcementMode    Name         PolicyDefinitionId
            ResourceGroup    Scope
-------------------  -------------  -----------------  -----------  ----------------------------------------------------------------------------------------------------------------------------------
----------  ---------------  ------------------------------------------------------------------------------------------
1.*.*                             Default            ArcLinuxAMA  /subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/providers/Microsoft.Authorization/policyDefinitions/A
rcLinuxAMA  ArcProjectRG     /subscriptions/afd82d75-3dfb-4266-badc-e0c5afd8b8d1/resourceGroups/ArcProjectRG
PS C:\Users\Msi\Desktop\AzureArc>
```

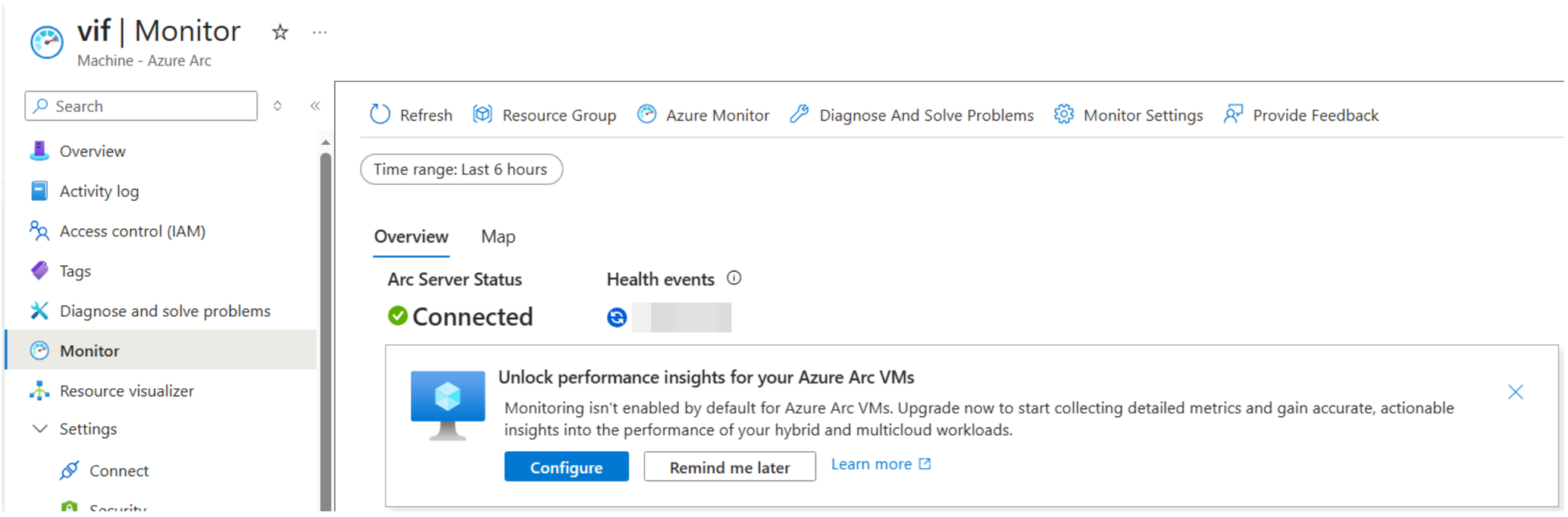
2 - Enabling Azure Extensions for Monitoring



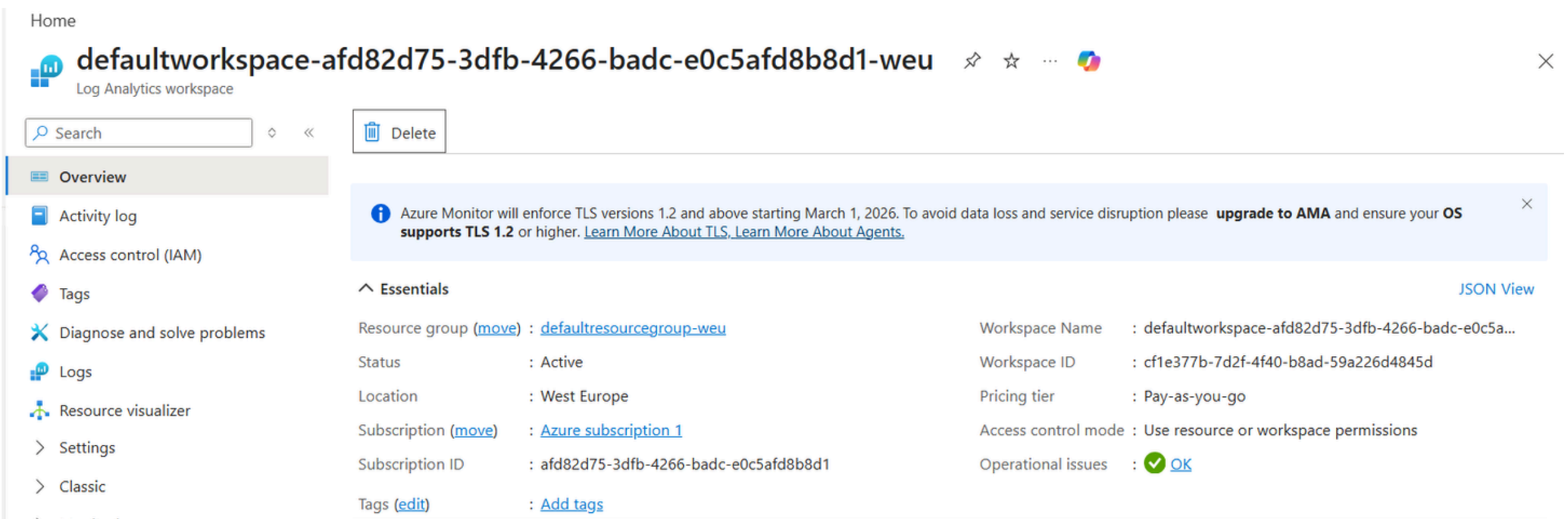
Installing Azure Monitor Agent for Linux Extension







Creating a Log Analytics Workspace



Creating a Data Collection Rule

Home > defaultworkspace-afd82d75-3dfb-4266-badc-e0c5afd8b8d1-weu

Create Data Collection Rule

Data collection rule management

Basics | Resources | Collect and deliver | Tags | Review + create

Pick a set of resources to collect data from. The Azure Monitor Agent will be automatically installed on virtual machines, scale sets, and Arc-enabled servers. For AKS clusters, managed Prometheus will automatically be enabled.
For Windows 10 and 11 devices, download the client installer and follow the guidance

This will also enable System Assigned Managed Identity on these resources, in addition to existing User Assigned Identities (if any).

+ Add resources + Create endpoint

Enable Data Collection Endpoints

Only resources in the same region can be assigned to the same endpoint. Learn more

Name	Type	Location	Resource group	Subscription
vif	Machine - Azure Arc	West Europe	ArcProjectRG	Azure subscription 1

Showing 1 - 1 of 1 results.

Review + create | < Previous | Next : Collect and deliver >

Configuring the DCR to collect Performance Counters

Basics | Resources | Collect and deliver | Tags | Review + create

Configure which data sources to collect and where to send the data to.

+ Add data source

Data source	Destination(s)
Performance Counters	Azure Monitor Metrics (preview)

Running KQL queries to monitor the VM

defaultworkspace-afd82d75-3dfb-4266-badc-e0c5afd8b8d1-weu | Logs

Log Analytics workspace

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Logs
Resource visualizer
Settings
Classic
Monitoring
Automation
Help

New ... | Observability agent New | Queries hub

Run | Time range : Set in query | Show : 1000 results | KQL mode

```
Heartbeat
| where TimeGenerated > ago(24h)
| summarize count() by Computer
```

Results | Chart

Computer	count_
vif	22

Columns

VI - Repository Publishing

1 - Creating a .gitignore file

```
.gitignore
1   terraform.tfvars
2   *.tfstate
3   *.tfstate.backup
4   .terraform/
```

2 - Pushing the files to the GitHub repository

```
PS C:\Users\Msi\Desktop\AzureArc> git init
Initialized empty Git repository in C:/Users/Msi/Desktop/AzureArc/.git/
PS C:\Users\Msi\Desktop\AzureArc> git add .
warning: in the working copy of '.terraform.lock.hcl', LF will be replaced by CRLF the next time Git touches it
```

```
PS C:\Users\Msi\Desktop\AzureArc> git commit -m "Initial commit"
[master (root-commit) c3a0908] Initial commit
 7 files changed, 142 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 .terraform.lock.hcl
 create mode 100644 main.tf
 create mode 100644 outputs.tf
 create mode 100644 policy.tf
 create mode 100644 provider.tf
 create mode 100644 variables.tf
PS C:\Users\Msi\Desktop\AzureArc>
```

```
PS C:\Users\Msi\Desktop\AzureArc> git remote add origin https://github.com/ayoubanlouf/HybridCloudGovernance
PS C:\Users\Msi\Desktop\AzureArc> git branch -M main
PS C:\Users\Msi\Desktop\AzureArc> git push -u origin main
```

```
PS C:\Users\Msi\Desktop\AzureArc> git push --set-upstream origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (11/11), 3.06 KiB | 1.02 MiB/s, done.
Total 11 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/ayoubanlouf/HybridCloudGovernance
   87d67fd..380a1e7  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Msi\Desktop\AzureArc>
```